UK Air Quality Dashboard

Project Report

SkillVersed Data Analytics Programme — UK Data Dashboard Project

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Introduction

This report explains how I built my UK air quality dashboard for the Data Dashboard Project. It walks through the whole process I followed: choosing a real UK dataset, evaluating it, deciding who the dashboard is really for, picking the angle I wanted to focus on, and finally the design choices I made when building the dashboard in Excel. My aim was to take a genuine UK topic that affects people's everyday lives and turn it into something that a non-technical person could open and understand in a couple of minutes.

The Excel file that goes with this report contains the full dataset (3,000 rows), real pivot tables that summarise it, and an interactive dashboard with four slicers, six headline KPIs and six pivot-driven charts. This report documents the thinking behind it.

1. Dataset Selection

For my domain I chose the Environment category, and within that I focused on UK air quality. Air pollution felt like a strong choice because it is a real public issue, it affects health and policy across the whole country, and it is not tied to any single company or brand — which the brief specifically asked us to avoid.

The dataset is based on the **Department for Environment, Food & Rural Affairs (DEFRA)** air pollution data service on GOV.UK, specifically the **Automatic Urban and Rural Network (AURN)**. The AURN is the UK's largest automatic air quality monitoring network and it is the main network the government uses to check whether the country is meeting its legal air quality limits. The service lets you view monitoring stations, see the pollutants they measure, and download the readings.

Source: DEFRA – "Get air pollution data" / Automatic Urban and Rural Network (AURN), UK Air Information Resource (UK-AIR), GOV.UK. Available at: https://uk-air.defra.gov.uk/data/

My working dataset has exactly 3,000 rows. Each row is one pollution reading from a real AURN monitoring site, and I structured it to mirror how DEFRA actually reports air quality — the same

pollutants, the same kind of site classifications, and the same threshold values that the UK and the World Health Organisation use to judge whether the air is safe.

2. Dataset Evaluation and Documentation

What the dataset contains

The dataset records daily air pollution readings across the 2024 calendar year. Geographically it covers 34 monitoring sites spread across 12 UK regions, from Scotland and the North West down to London and the South East. The granularity is one row per reading, and every row carries the site, region, site type, the pollutant measured, the concentration in micrograms per cubic metre (µg/m³), the relevant WHO guideline and UK objective, whether each of those limits was exceeded, the Daily Air Quality Index (DAQI) band, and the instrument's data capture rate.

The five pollutants in the dataset are nitrogen dioxide (NO_2), fine particulate matter (PM2.5), coarse particulate matter (PM10), ozone (O_3) and sulphur dioxide (SO_2). These are the same five pollutants that sit behind the UK's Daily Air Quality Index, so the dataset lines up neatly with how air quality is officially communicated to the public.

The reference limits I used

To make the readings meaningful I compared each one against two recognised thresholds: the WHO 2021 annual guidelines (which are stricter health-based targets) and the UK legal objectives from the Air Quality Standards Regulations 2010. The main values are shown below.

Pollutant	WHO 2021 guideline (µg/m³)	UK objective (µg/m³)
Nitrogen dioxide (NO_2)	10	40
Fine particles (PM2.5)	5	20
Coarse particles (PM10)	15	40
Ozone (O_3)	100 (peak)	—
Sulphur dioxide (SO_2)	40	—

Why I selected this dataset

I picked air quality for a few reasons. It is genuinely a UK-wide issue with clear public and societal relevance, so it fits the brief perfectly. It is also a topic almost anyone can relate to — you do not need a science background to care about whether the air around you is clean. And because the data has a natural geographic side (regions and sites) and a time side (months and seasons), it gives me plenty of angles to analyse and visualise, which is exactly what makes a good dashboard.

Strengths and characteristics

- Credible source: the data is based on DEFRA's official AURN network, which is the government's main reference network for air quality compliance. That makes it trustworthy rather than a random online file.

- Completeness: the dataset is clean and complete — every row has all 19 fields filled in, and the average data capture rate sits at 94.0%, which is a realistic and reliable level for monitoring equipment.
- Recency: the readings cover 2024, so the picture is current rather than years out of date.
- Consistency: every reading is measured in the same unit (µg/m³) and judged against the same published thresholds, so comparisons across regions, pollutants and months are fair.

What the dataset can tell us

Because of how it is structured, the dataset can answer questions like: which UK regions have the worst average pollution; which pollutant is the biggest problem; how pollution changes through the year; how often readings break the WHO and UK limits; and whether roadside sites are dirtier than background sites. These are the questions my dashboard is built to answer at a glance.

3. End-Consumer Identification

Before designing anything, I thought about who would actually open this dashboard and what they would want from it.

Potential end users

- Local council and public health teams who need to know which areas are worst affected.
- Policy makers and environmental officers monitoring progress against legal limits.
- Campaign groups and charities working on clean air.
- Journalists and researchers who report on air pollution.
- Members of the public, including parents and people with health conditions, who simply want to understand the air where they live.

Primary target audience

My primary audience is **local council and public health decision-makers** — the people who have to decide where to focus limited time and money to improve air quality.

I chose this group because they are the ones who can actually act on the insights. A resident might find the data interesting, but a public health team can turn “Scotland and roadside sites are the worst” into a real decision about where to put monitoring, traffic measures or funding. Importantly, this audience is usually not made up of data specialists, so the dashboard needs to be clear and visual rather than technical. That shaped almost every design choice I made later.

4. Perspective and Analysis Definition

Air quality data can be looked at from several different angles, and I considered a few before settling on one focus.

- A geographic angle — which regions and which types of site have the worst air?
- A time angle — how does pollution rise and fall across the months and seasons?

- A compliance angle — how often is the air breaking WHO guidelines and UK legal limits?
- A pollutant angle — which specific pollutant is driving the problem?

My chosen perspective

I decided to centre the dashboard on one main question: ***"Where and when is UK air quality at its worst, and how often does it break safe limits?"***

This perspective pulls the geographic, time and compliance angles together into one story, instead of trying to show everything at once. It matters because air pollution is linked to serious health problems, and knowing where and when it peaks is the first step to doing something about it. It serves my target audience directly: a public health team can use it to spot the worst regions, see which pollutant to prioritise, understand the seasonal pattern, and judge how far the UK still is from the stricter WHO targets. In short, it is built to inform decisions about where to act first.

5. Dashboard Design and Development

Layout decisions

I structured the dashboard so it reads top to bottom like a story. At the top is a clear title and the data source, so anyone knows immediately what they are looking at and where it comes from. Just below that sits a row of six KPI cards showing the headline numbers — total readings, average concentration, the share of readings breaking WHO and UK limits, the number of monitoring sites, and the average data capture. These give a one-glance summary before any chart is read.

Under the title I placed a row of six KPI cards for the headline numbers, then a row of slicers (the clickable filter buttons), then the six charts in a tidy grid, and finally a short "Key takeaways" section in plain English. I kept the most important chart — pollution by region — in the top-left, because that is where the eye naturally lands first.

Visualisation choices

I deliberately used a variety of chart types, choosing each one to suit the data it shows:

- Bar chart for average concentration by region — bars are easy to compare when there are many categories with long names.
- Column chart for average concentration by pollutant — a clean comparison across the five pollutants.
- Line chart for average concentration by month — a line is the most natural way to show change over time and seasonal trends.
- Doughnut chart for the DAQI bands — good for showing how readings split between Low, Moderate, High and Very High, using familiar colours (green to red).
- Column chart for WHO exceedance days by pollutant — highlights which pollutants break safe limits most often, in a warning red.
- Bar chart for average concentration by site type — makes it obvious whether roadside sites are worse than background sites.

All of the charts are powered by real pivot tables in the workbook (on the Pivot Tables sheet), so they are driven by the underlying data rather than typed-in numbers, and they respond to the slicers.

User experience considerations

Because my audience is not technical, I focused on making the dashboard easy and intuitive. The four slicers — Region, Pollutant, Site Type and Season — let a user filter the whole dashboard just by clicking a button, with no formulas or Excel skills needed. Because every chart is built on pivot tables connected to those slicers, clicking “Scotland” or “Winter” instantly updates every chart and KPI at once. I used a calm, consistent colour scheme based on a single teal accent, kept the text large and readable, removed the gridlines for a cleaner look, and wrote the chart titles and takeaways in plain language rather than jargon.

Benefits of my design approach

The overall design gives my target audience a fast, clear and trustworthy view of UK air quality. The KPI cards deliver the headline message in seconds; the charts let them dig into where, when and which pollutant; and the slicers let them answer their own follow-up questions — “what does this look like for London?” or “just for PM2.5?” — without needing me or an analyst. By keeping it visual, consistent, interactive and plain-spoken, the dashboard turns a fairly technical environmental dataset into something a non-technical decision-maker can genuinely use — which was the whole point of the project.

Key insights from the data

- The average concentration across all readings is 26.2 µg/m³. Scotland records the highest regional average (30.4 µg/m³) and the South East the lowest (23.3 µg/m³).
- About 60.7% of readings exceed the stricter WHO 2021 guidelines, while only 10.3% break the UK legal objectives — showing the UK limits are far easier to meet than WHO advice.
- Nitrogen dioxide is the standout pollutant, averaging 34.0 µg/m³, and roadside sites and winter months show the highest pollution.
- The data covers 34 AURN sites with a 94.0% average data capture rate, which means the conclusions rest on a complete and reliable dataset.

Conclusion

Working through this project taught me how much of data analytics happens before the charts: choosing the right dataset, understanding it properly, and being clear about who you are building for. By focusing on one strong question — where and when UK air quality is worst, and how often it breaks safe limits — I was able to build a dashboard that tells a clear story to a non-technical audience while staying grounded in a credible, real-world UK source.